Belmont City College

MULTIPLE CHOICE ANSWER SHEET

CHEMISTRY 12 SEMESTER 2 EXAMINATION 2013

Name:____________________________**Teacher:** _______________________________

For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. For example, if b is your answer: □a ■ b □c □d

If you make a mistake, place a cross through that square, do not erase or use correction fluid.
Shade your new answer.

For example, if b is a mistake and d is your correct answer: □a ☒b □c ■d

In the event that you then change your mind back to your original answer, you then cross out the second selection and then circle the first choice.

For example, if b was the first choice and d your second, but you changed your mind back, and b is your correct answer:

□a ☒b □c ☒d

Multiple Choice: Mark the answer by shading in the box to the LEFT of the letter you chose.

1. □a □b □c □d
2. □a □b □c □d
3. □a □b □c □d
4. □a □b □c □d
5. □a □b □c □d
6. □a □b □c □d
7. □a □b □c □d
8. □a □b □c □d
9. □a □b □c □d
10. □a □b □c □d
11. □a □b □c □d
12. □a □b □c □d
13. □a □b □c □d
14. □a □b □c □d
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19. □a □b □c □d
20. □a □b □c □d
21. □a □b □c □d
22. □a □b □c □d
23. □a □b □c □d
24. □a □b □c □d
25. □a □b □c □d

Section Two: Short answer 35% (70 Marks)

This section has **eight (8)** questions. Answer **all** questions. Write your answers in the space provided.

When calculating numerical answers, show your working or reasoning clearly. **Express numerical answers to three significant figures and include appropriate units** where applicable.

You can use abbreviations, such as **'nvr'** for 'no visible reaction'.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

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- Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Suggested working time: 60 minutes.

Question 26 **(8 marks)**

For each of the following pairs of substances, provide details of a **chemical** test that would allow the two substances to be distinguished from one another. Equations are **not** required.

substances	chemical test	Observations
Ni(s) and Mg(s)		**Ni(s)**
		Mg(s)
$NaCH_3COO(s)$ and $NaC\ell(s)$		**$NaCH_3COO(s)$**
		$NaC\ell(s)$

Question 27 **(8 marks)**

(a) 2-methylpropanal, whose formula is $(CH_3)_2CHCHO$, has many structural isomers. In the spaces provided below, draw the full structure and give the IUPAC names of two isomers.
(4 marks)

Structure	IUPAC name

2-methylpropanal can be converted into substance Y by heating it with a mixture of sodium dichromate and dilute sulfuric acid.

(b) State an observation that can be made as this reaction proceeds. (1 mark)

____

____

(c) Name the functional group present in substance Y that is **not** present in 2-methylpropanal.
(1 mark)

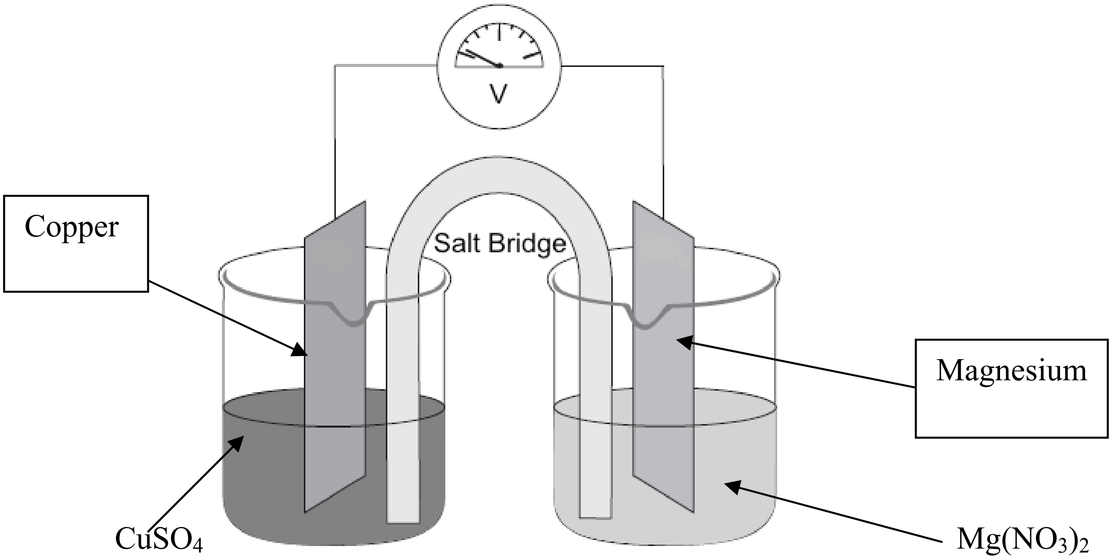
____

(d) Write a balanced half-equation showing the conversion of 2-methylpropanal into substance Y
(2 marks)

____

Question 28 **(11 marks)**

A student wishing to investigate the oxidising strength of various metals set up an electrochemical cell made up of a copper rod immersed in a solution of copper (II) sulfate and a magnesium rod immersed in a solution of magnesium nitrate. The apparatus used is shown in the diagram below.



a) Clearly label the anode on the diagram above. (1 mark)

b) By adding an **arrow** to the diagram, show the direction of electron flow in the external circuit. (1 mark)

c) Give the half equation for the reaction occurring at the positive electrode. (1 mark)

____

d) Give the name or formula of a suitable substance that the salt bridge might contain. (1 mark)

____

e) State the reducing agent (reductant) in the cell. (1 mark)

____

Question 28 (contd)

f) The initial voltage measured in the cell was higher than the 2.70 V that the student had expected to measure. Give **two** possible reason for this observation. (2 marks)

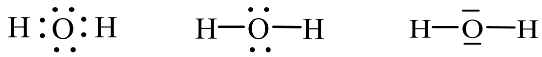
g) Describe two observation that would be expected to be made in the copper half-cell whilst the experiment was taking place. (2 marks)

Rusting can be described as the formation of iron (III) hydroxide from iron in the presence of water and oxygen. Rusting can be prevented by attaching magnesium to the object that is to be protected.

h) With reference to the list of redox potentials found on your data sheet, explain why magnesium is a suitable material for this purpose. (2 marks)

Question 29 **(9 marks)**

For each species listed in the table below, draw the Lewis structure, representing all valence shell electron pairs either as : or as — **and** state or sketch the shape of the species.

(for example, water)  (bent, polar)

Species	Lewis structure (showing all valence electrons)	Shape (sketch or name)
phosphorus trichloride **$PC\ell_3$**		
methanal **HCHO**		
nitrate ion **NO_3^-**		

Question 30 **(10 marks)**

By referring to the structure and/or bonding present, account for the following:

(a) Soaps can dissolve both polar and non polar solutes

(4 marks)

(b) Aluminium is a better conductor of electricity than magnesium. (2 marks)

(c) Silicon dioxide (SiO_2) is a solid with an extremely high melting point, whilst carbon dioxide (CO_2) sublimes at -78°C. (4 marks)

Question 31 **(7 marks)**

(a) But-1-ene and but-2-ene are isomers of one another. What is the name given to this type of isomerism? (1 mark)

____

(b) In addition to the type of isomerism displayed by the molecules in part (a), but-2-ene itself exhibits another type of isomerism. State the name given to this type of isomerism, and identify the features of but-2-ene that make this type of isomerism possible. (3 marks)

____

____

____

(c) In the space provided, draw a length of polymer chain that could be formed from 2-butene, showing **three** repeating units. (2 marks)

(d) State the name given to the type of polymerisation described in part (c). (1 mark)

____

Question 32 **(7 marks)**

(a) Nitrous acid (HNO_2) is a weak acid. Use relevant equations to explain how you would expect the pH of a 0.1 mol L^{-1} solution of nitrous acid to compare with that of a 0.1 mol L^{-1} solution of nitric acid. (3marks)

A titration is carried out using 25.00 mL of approximately 0.9 molL^{-1} ammonia solution placed in a conical flask. A few drops of indicator are added, and 0.900 mol L^{-1}nitric acid is added from a burette.

The table below shows some indicators, together with their pH ranges, and their colours.

indicator name	pH range	colour in acid	Colour in base
methyl yellow	2.9 – 4.0	Red	Yellow
bromothymol blue	6.0 – 7.6	Yellow	Blue
nile blue	10.1 – 11.1	Blue	Red
Nitramine	11.0 – 13.0	Colourless	Orange

(b) From the indicators shown in the table, choose one that would be suitable for the titration, and state the colour change that would be expected to occur. (2 marks)

Indicator: ______________________________

Colour change: ______________________________

When approximately 12.5 mL of nitric acid have been added, the solution in the flask can be classified as a buffer solution.

(c) With the aid of equations, explain how the solution is able to act as a buffer. (2 marks)

Question 33 **(10 marks)**

When ethanol and ethanoic acid are mixed together in concentrated sulfuric acid, the following reversible reaction occurs:

$$CH_3CH_2OH\ (aq) + CH_3COOH\ (aq) \rightleftharpoons CH_3CH_2OOCCH_3\ (aq) + H_2O\ (\ell)$$

(a) Identify the type of reaction taking place in the forward direction, and give the IUPAC name of the organic product. (2 marks)

Type of reaction: ____________________

Name of product: ____________________

(b) Write an expression for the equilibrium constant, K, for the reaction in the space provided. (1 mark)

(c) The reaction is usually carried out at high temperatures in order to increase the yield. State and explain what this tells us about the enthalpy (H_p) of the products compared to the enthalpy (H_R)of the reactants. (3 marks)

Question 33 (cont'd)

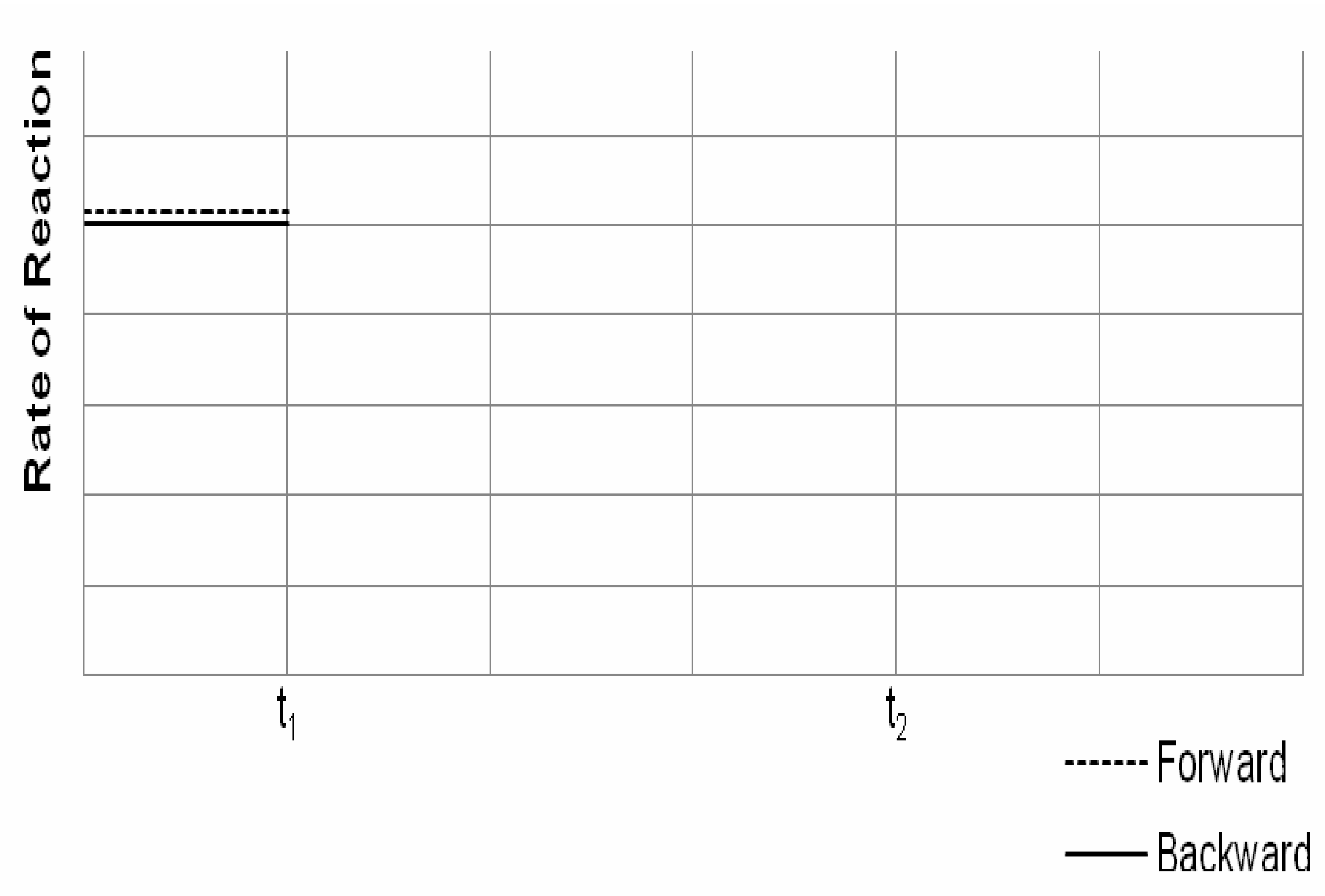
A titration can be used to determine the proportions of reactants and products present at equilibrium.

Although the addition of sodium hydroxide does disturb the equilibrium, the inaccuracy introduced is small if the reaction mixture is cooled immediately prior to titration.

(d) State the effect of addition of sodium hydroxide to the equilibrium mixture on the value of the **equilibrium constant**, K, by circling the correct answer below. (1 mark)

Effect on K (circle one)	INCREASE	DECREASE	NO CHANGE

(e) On the axes shown below sketch the effect of cooling the reaction mixture (t_1) on the rates of the forward and backward reactions until the system returns to a new equilibrium (t_2).(3 marks)



End of section two

Section Three: Extended answer 40% (80 marks)

This section contains **six(6)** questions. Answer **all** questions. Write your answers in the spaces provided.

Where questions require an explanation and/or description, marks are awarded for the relevant chemical content and also for coherence and clarity of expression.

Final answers to calculations should be expressed to three (3) significant figures.

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Suggested working time:70 minutes.

Question 34 **(13 marks)**

Chlorine is found in acids of various strengths. Chlorine can exist in different oxidations states. Some acids are shown in the table below.

Name and formula of acid	Strong/Weak
hydrochloric acid $HC\ell$	Strong
hypochlorous acid $HOC\ell$	Weak
chlorous acid $HC\ell O_2$	Weak
chloric acid $HC\ell O_3$	Strong
perchloric acid $HC\ell O_4$	Strong

(a) What mass of perchloric acid would need to be dissolved in 250 mL of distilled water to produce a solution with a pH of 3.59? (3 marks)

2.78 g of solid barium hydroxide was added to a beaker containing 200 mL of a 0.160 mol L^{-1} aqueous solution of chloric ($HClO_3$) acid, and the solution stirred until all the solid had dissolved. The product formed is soluble.

(b) Write a balanced ionic equation to show the reaction taking place in the beaker. (2 marks)

(c) Determine the limiting reagent by calculation. (4 marks)

(d) Calculate the pH of the resulting solution. (4 marks)

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Question 35 **(18 marks)**

Solutions of iron(II) salts are often used in redox titrations, but can be problematic as the Fe^{2+} ions can be oxidised by oxygen in the environment, forming various hydrated forms of iron(III) oxide, according to the following rusting equation:

$$Fe^{2+}(aq) + O_2(g) + (2x + 4)\, H_2O(\ell) \rightleftharpoons 2\, Fe_2O_3.x\, H_2O(s) + 8H^+(aq)$$

Ammonium iron (II) sulfate, or Mohr's salt, is often preferred over iron(II) sulfate for redox titration purposes since the unwanted oxidation of Fe^{2+} is prevented by the ammonium ions present (which also reduces the pH of the solution). Mohr's salt is commonly found in hydrated form, as any of a number of salts with the formula $(NH_4)_2Fe\,(SO_4)_2.x\, H_2O$.

(a) Write a hydrolysis equation to show how the ammonium ions are able to lower the pH of the solution. (2 marks)

___

(b) Use the rusting equilibrium reaction above to explain why the oxidation of Fe^{2+} is prevented in solutions of low pH. (2 marks)

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___

___

10.0 g of hydrated ammonium iron(II) sulfate crystals were dissolved in distilled water and made up to 250.0 mL in a volumetric flask.
25.0 mL aliquots of this solution were titrated against 0.0240 mol L^{-1} potassium permanganate solution until consistent results were obtained. The table below shows the results of the experiment.

	Rough	**1**	**2**	**3**	**4**
Final volume (mL)	23.00	21.25	21.25	22.65	23.35
Initial volume (mL)	0.00	0.05	0.00	0.10	2.10
Titre or Titration volume (mL)					

(c) Complete the table and calculate the average titration volume. (1 mark)

Average titre: ___

Question 35 (continued)

(d) Write an overall (net) balanced ionic equation for the titration reaction taking place.
(3 marks)

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___

___

(e) Calculate the number of moles Fe^{2+} present in the 250.0 mL volumetric flask.

(4 marks)

(f) Determine the mass of non-hydratedMohr's salt present in 250.0 mL (3 marks)

(g) Calculate the value of x in the formula $(NH_4)_2Fe(SO_4)_2.x\ H_2O$. (3 marks)

Question 36 **(10 marks)**

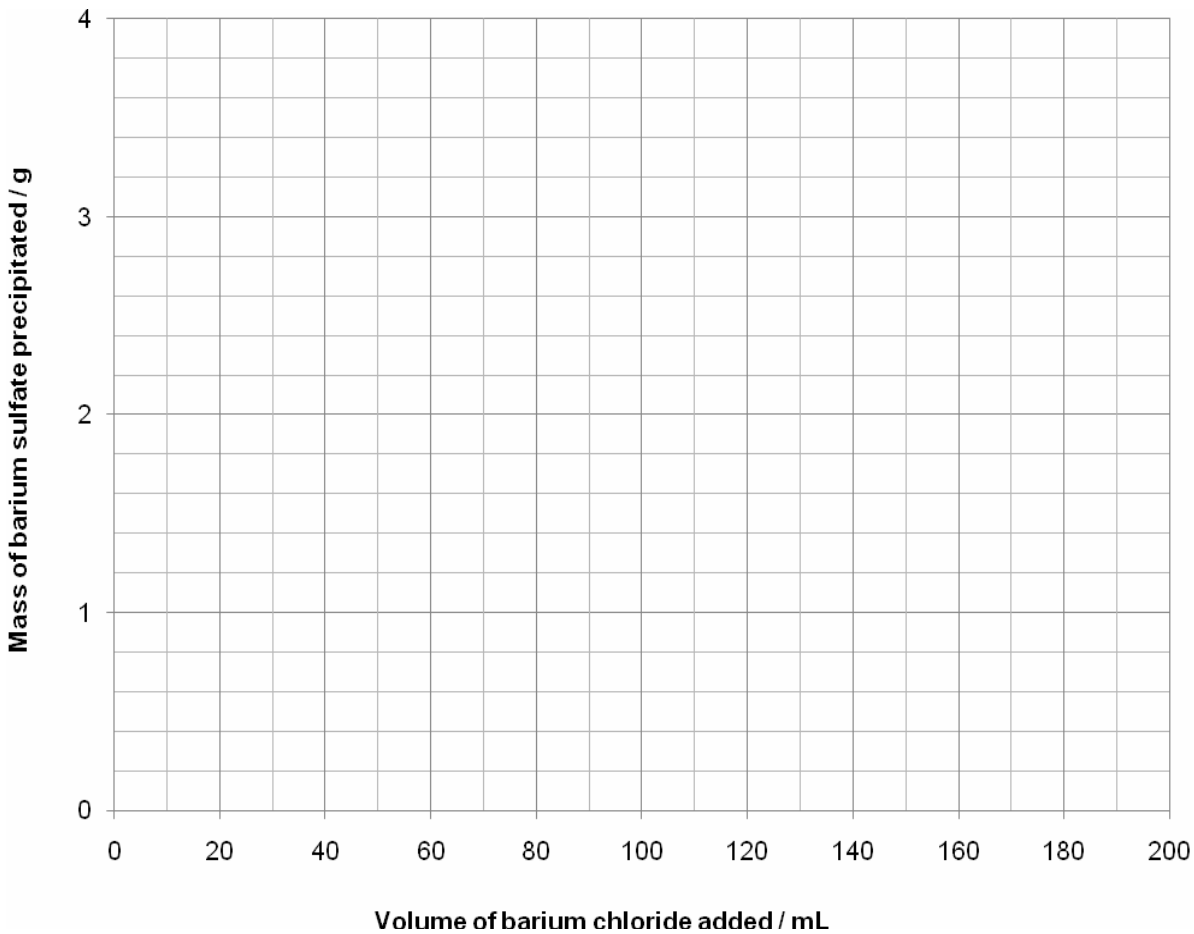
A sample of powdered magnesium sulfate is known to have been contaminated with sodium chloride. The percentage purity of the magnesium sulfate can be determined by the following method:

- 32.50 g of the impure magnesium sulfate is dissolved in water and the solution is made up to 500.0 mL in a volumetric flask.
- Six 20.0 mL aliquots of this solution are placed in separate conical flasks.
- Different volumes of 0.100 mol L^{-1} $BaC\ell_2(aq)$ are added to each flask, causing any sulfate ion present to precipitate out of the solution.
- The precipitate from each sample is filtered, rinsed with distilled water and then dried to constant mass.

The results of this analysis are shown in the table below.

Sample	**1**	**2**	**3**	**4**	**5**	**6**
Volume of $BaC\ell_2(aq)$ added(mL)	30.0	60.0	90.0	120.0	150.0	180.0
Mass of $BaSO_4(s)$ precipitated(g)	0.61	1.23	1.83	2.04	2.04	2.04

(a) Graph the results in a suitable format using the axes provided. (2 marks)



Question 36

(b) Write a balanced ionic equation for the reaction taking place. (1 mark)

(c) Explain why the mass of precipitate remained constant for the last three samples, in spite of the fact that more barium chloride was being added. (1 mark)

(d)Use the graph you have drawn in part (a) to estimate the <u>minimum</u> volume of barium chloride needed to precipitate all the sulfate ions from solution. (1 mark)

(e) Use the mass of precipitate to calculate the percentage purity of the magnesium sulfate. (5 marks)

Question 37 **(12 marks)**

Commercial caustic soda contains both sodium hydroxide and sodium carbonate. A year 12 student wants to determine the percentage by mass of sodium carbonate in caustic soda. She dissolves 2.487gof the solid, transfers it carefully to a 250.0 mL volumetric flask, and adds distilled water to the mark.

She then pipettes 25.0 mL of this solution into a conical flask and titrates it against standard 0.9331mol L^{-1}hydrochloric acid using methyl orange indicator. The average volume of the acid added from the burette was 21.34 mL. In this case hydrochloric acid reacts with all the carbonate and hydroxide ions present in the conical flask.

She pipettes another 25.0 mL of the diluted caustic solution from the volumetric flask and places it into another separate conical flask. She warms the conical flask and its contents to 70°C and slowly adds 1%barium chloride solution until no more precipitate is formed. This process removes all the carbonate ions by precipitating them out as barium carbonate. She cools the solution, adds a few drops of phenolphthalein indicator to it and titrates it against the 0.9331 mol L^{-1}hydrochloric acid. The average volume of hydrochloric acid from the burette was 19.82 mL. In this case the hydrochloric acid only reacts with hydroxide ions in solution.

Show **all** working out steps clearly to score full allocated marks. Express your answer correct to three significant figures.

(a) The number of moles of hydrochloric acid that would be required to neutralise all of the carbonate and hydroxide ions present in 250.0 mL volumetric flask. (2 marks)

(b) The number of moles of sodium hydroxide present in 250.0 mL volumetric flask. (3 marks)

(c) The mass of sodium carbonate present in 250.0 mL solution. (5 marks)

(d) What percentage by mass of sodium carbonate in the caustic soda will her results give? (2 marks)

Question 38 **(10 marks)**

The functional groups present in organic molecules can, by definition, have a strong influence on the chemical properties of those molecules, but they can also play a role in determining the physical properties of substances.

The table below outlines some of the physical properties of four organic substances; pentane, 2,2-dimethylpropane, pent-2-ene and propanoic acid.

	Boiling point (°C)	**solubility in water**
pentane	36.1	Low
2,2-dimethylpropane	9.5	Low
pent-2-ene	37.0	Low
propanoic acid	144.1	High

With clear reference to the structure and bonding present, compare and contrast the **chemical and physical properties** of the four substances. Your answer should include discussion relating to data provided in the table above and common chemical reactions of the hydrocarbons and the functional group chemistry.

Your answer must include equations where appropriate.

Marks are awarded for clarity of communication. Answers may be written as a series of dot points and diagrams may be used, but care should be taken to ensure that there is a logical sequence of ideas and that any abbreviations or diagrams are explained clearly.

Question 39 **(17 marks)**

The European Aviation Safety Agency (EASA) have recently sponsored studies into ultra-low sulfur jet (ULSJ) fuel standard. It is estimated that, by reducing the amount of sulfur in jet fuel, between 1000 and 4000 pollution-related deaths could be prevented globally each year.

Sulfur is present in fossil fuels in the form of sulfur- containing organic compounds, such as dibenzothiophenes. Analysis of one such compound showed that it was made up only of the elements carbon, hydrogen and sulfur. Upon combustion in excess oxygen, a 22.60 g sample of the compound was found to produce 8.85 g of water vapour and 77.9 L of carbon dioxide, measured at 1000°C and
200 kPa.

(a) Calculate the **empirical formula** of the compound. (8 marks)

Question 39 (continued)

Another sample of the compound, weighing 10.71g, was vapourised in the absence of oxygen. The vapours occupied 1.265 L at 200 kPa and 250°C.

(b) Use this information and your answer to part (a) to calculate the **molecular formula** of the compound. (3 marks)

Question 39 (continued)

The ULSJ fuel standard is equivalent to a concentration of sulfur 15 ppm (parts per million).

(c) Calculate the concentration of sulfur in ULSJ fuel in mol L^{-1} if1L of the fuel weighs 800 g.
(3 marks)

One of the problems associated with the presence of sulfur in fuels is that as rain falls through oxides of sulfur (SO_2) in the atmosphere, they react with the rain. The effect is that the rain becomes acidic.

(d) With the help of equations, explain how these oxides can cause rainwater to become acidic.
(3 marks)

End of semester 2 chemistry exam and questions

Spare answer page

Question number: _______________

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Question number: ______________

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Question number: _______________

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Question number: _______________